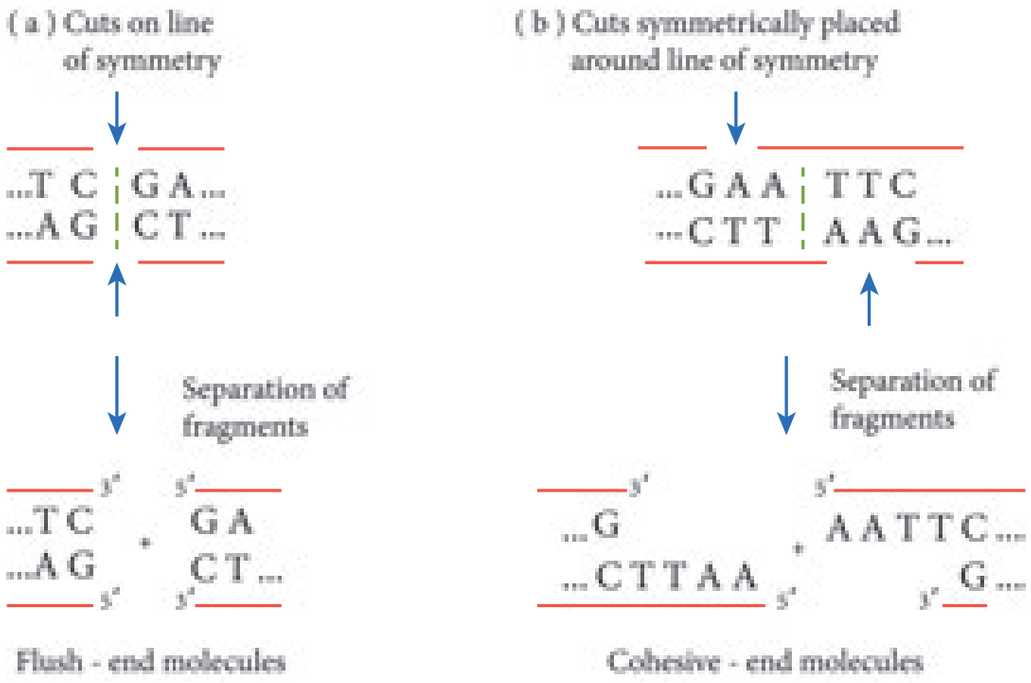


Figure 12.29: Two types of cuts made by restriction enzymes

two single – stand breaks) one break in each strand. There are two distinct arrangements of these breaks 1. both breaks at the center of symmetry (generating flush or blunt ends) or 2. breaks that are symmetrically placed around the line of symmetry generating cohesive ends. Figure 12.29 shows two types of cuts made by restriction enzymes. The arrow indicates the cleavage site. The dashed line is the center of symmetry of the sequence (Table 12.2).

Application of Recombinant DNA Technology

a. Production of medically useful proteins such as somatostain, insulin, human growth hormone and interferon. It decreases the dependency on human tissues and solves problem of limited production.

b. Development of synthetic vaccines for instance, vaccines for malaria and rabies a recombinant hepatitis vaccine is already commercially available.

Type	No of Enzyme and sub units	Cleavage site	Examples	Bacterial source
I	One with 3 sub units for recognition cleavage and methylation	1000 bp from recognition site	EcoK1 Cfr A1	*Escherichia coli* *Citrobacter fruendii*
II	Two different enzymes to cleave or modify the recognition sequence	Same as recognition or close to recognition site	Eco R1 Alu I	*Escherichia coli* *Arthrobacter luteus*
III	One with 2 subunits	24- 26 bp from recognition site	Hinf III Pst II	*Haemophilus influenzae* *Providencia stuarti*

Table 12.2: Types and features of restriction enzyme

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